

Architecture Justification

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Vaccination Cohort & Dose Scheduling System

We chose a **Microservices Architecture** for the Vaccination Cohort & Dose Scheduling System. The main reason is that the system has several different tasks, such as registration, checking dose eligibility, booking slots, generating certificates, and sending reminders. Instead of putting everything into one large application, these functions can be divided into separate services that communicate through well-defined interfaces.

Scalability and Availability

One of the main reasons for choosing microservices is the expected increase in users when a new vaccination group becomes eligible. The system should support up to **100,000 concurrent users** and maintain **99.9% availability**. With microservices, services that receive more traffic, such as the scheduling service, can be scaled independently without having to scale the whole system. If one service has a problem, the other services can also continue working, which improves overall reliability.

Separation of Responsibilities

Each service in the design has a specific responsibility. The **Cohort & Eligibility Service** checks whether a citizen is eligible for the next dose and makes sure the required interval has passed. The **Vaccination Scheduling Service** handles available slots and bookings. The **Certificate & QR Service** deals with vaccination certificates and QR verification, while the **Notification Service** handles reminders and booking confirmations. This makes the system easier to develop, test, and maintain because changes in one area do not require major changes to unrelated services.

Security

Security is important because the system stores vaccination records and generates digital certificates. Keeping certificate generation and QR verification in a separate service makes it easier to control access to this information. The service can also handle digital-signature verification without exposing the complete vaccination database to other components. The system also needs to support secure QR verification in both online and offline modes.

Performance

The architecture also helps with performance. QR verification has to be completed in **less than 150 ms**, so the Certificate & QR Service can be optimized and scaled separately when there are many verification requests. Similarly, the scheduling service can be scaled during periods when a large number of citizens are trying to book vaccination slots at the same time.

Conclusion

Overall, microservices are a good fit for this system because they provide **better scalability, separation of responsibilities, reliability, security, and performance**. The component diagram shows these separate services along with their provided and required interfaces, making it clear how the different parts of the vaccination system communicate with each other.